

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Numerical

Let $f : (0, 2) \rightarrow \mathbb{R}$ be defined as $f(x) = \log_2 \left(1 + \tan \left(\frac{\pi x}{4} \right) \right)$

Then, $\lim_{n \rightarrow \infty} \frac{2}{n} \left(f \left(\frac{1}{n} \right) + f \left(\frac{2}{n} \right) + \dots + f(1) \right)$ is equal to _____

Q2: 16 March (Shift 1) - Numerical

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that $f(x) + f(x+1) = 2$, for all $x \in \mathbb{R}$. If

$I_1 = \int_0^8 f(x) dx$ and $I_2 = \int_{-1}^3 f(x) dx$, then the value of $I_1 + 2I_2$ is equal to _____

Q3: 16 March (Shift 2) - Single Correct

Consider the integral

$$I = \int_0^{10} \frac{[x]e^{[x]}}{e^x - 1} dx$$

where $[x]$ denotes the greatest integer less than

or equal to x . Then the value of I is equal to:

(1) $9(e - 1)$

(2) $45(e + 1)$

(3) $45(e - 1)$

(4) $9(e + 1)$

Q4: 17 March (Shift 1) - Numerical

If $[\cdot]$ represents the greatest integer function,

then the value of

$$\left| \int_0^{\sqrt{\frac{\pi}{2}}} \left[[x^2] - \cos x \right] dx \right|$$

is _____

Questions with Answer Keys

MathonGo

Q5: 17 March (Shift 2) - Single Correct

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = e^{-x} \sin x$. If

$F : [0, 1] \rightarrow \mathbb{R}$ is a differentiable function

such that $F(x) = \int_0^x f(t)dt$, then the value of $\int_0^1 (F'(x) + f(x)) e^x dx$ lies in the interval

(1) $\left[\frac{327}{360}, \frac{329}{360} \right]$

(2) $\left[\frac{330}{360}, \frac{331}{360} \right]$

(3) $\left[\frac{331}{360}, \frac{334}{360} \right]$

(4) $\left[\frac{335}{360}, \frac{336}{360} \right]$

Q6: 17 March (Shift 2) - Single Correct

If the integral $\int_0^{10} \frac{[\sin 2\pi x]}{e^{x-[x]}} dx = \alpha e^{-1} + \beta e^{-\frac{1}{2}} + \gamma$,

where α, β, γ are integers and $[x]$ denotes the greatest integer less than or equal to x , then the value of

$\alpha + \beta + \gamma$ is equal to :

(1) 0

(2) 20

(3) 25

(4) 10

Q7: 17 March (Shift 2) - Numerical

Let $I_n = \int_1^e x^{19} (\log |x|)^n dx$, where $n \in \mathbb{N}$. If $(20) I_{10} = \alpha I_9 + \beta I_8$, for natural numbers α and β , then $\alpha - \beta$ equal to ____.

Q8: 18 March (Shift 1) - Numerical

Let $f(x)$ and $g(x)$ be two functions satisfying $f(x^2) + g(4-x) = 4x^3$ and $g(4-x) + g(x) = 0$, then the value of $\int_{-4}^4 f(x)^2 dx$ is

Q9: 18 March (Shift 2) - Single Correct

Questions with Answer Keys

MathonGo

Let $g(x) = \int_0^x f(t)dt$, where f is continuous function in $[0,3]$ such that $\frac{1}{3} \leq f(t) \leq 1$ for all $t \in [0, 1]$ and

$0 \leq f(t) \leq \frac{1}{2}$ for all $t \in (1, 3]$ The largest possible interval in which $g(3)$ lies is:

(1) $\left[-1, -\frac{1}{2}\right]$

(2) $\left[-\frac{3}{2}, -1\right]$

(3) $\left[\frac{1}{3}, 2\right]$

(4) $[1,3]$

www.mathongo.com

Q1 (1)

$$E = 2 \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} f\left(\frac{r}{n}\right)$$

$$E = \frac{2}{\ln 2} \int_0^1 \ln\left(1 + \tan \frac{\pi x}{4}\right) dx$$

replacing $x \rightarrow 1 - x$

$$E = \frac{2}{\ln 2} \int_0^1 \ln\left(1 + \tan \frac{\pi}{4}(1 - x)\right) dx$$

$$E = \frac{2}{\ln 2} \int_0^1 \ln\left(1 + \tan\left(\frac{\pi}{4} - \frac{\pi}{4}x\right)\right) dx$$

$$E = \frac{2}{\ln 2} \int_0^1 \ln\left(1 + \frac{1 + \tan \frac{\pi}{4}x}{1 + \tan \frac{\pi}{4}x}\right) dx$$

$$E = \frac{2}{\ln 2} \int_0^1 \ln\left(\frac{2}{1 + \tan \frac{\pi}{4}x}\right) dx$$

$$E = \frac{2}{\ln 2} \int_0^1 \left(\ln 2 - \ln\left(1 + \tan \frac{\pi x}{4}\right)\right) dx$$

equation

(i) + (ii)

$$E = 1$$

Q2 (16)

$$f(x) + f(x + 1) = 2$$

$\Rightarrow f(x)$ is periodic with period = 2

$$I_1 = \int_0^8 f(x) dx = 4 \int_0^2 f(x) dx$$

$$= 4 \int_0^1 (f(x) + f(1 + x)) dx = 8$$

Similarly $I_2 = 2 \times 2 = 4$

$$I_1 + 2I_2 = 16$$

Q3 (3)

$$I = \int_0^{10} [x] \cdot e^{[x]-x+1} dx$$

$$I = \int_0^1 0 dx + \int_1^2 1 \cdot e^{2-x} + \int_2^3 2 \cdot e^{3-x} + \dots + \int_9^{10} 9 \cdot e^{10-x} dx$$

$$\Rightarrow I = \sum_{n=0}^9 \int_n^{n+1} n \cdot e^{n+1-x} dx$$

$$= - \sum_{n=0}^9 n(e^{n+1-x})_n^{n+1}$$

$$= (e - 1) \sum_{n=0}^9 n$$

Hints and Solutions

MathonGo

$$= -\sum_{n=0}^9 n \cdot (e^0 - e^1)$$

$$= 45(e - 1)$$

Q4 (1)

$$I = \int_0^{\sqrt{\pi/2}} ([x^2] + [-\cos x]) dx$$

$$= \int_0^1 0 dx + \int_1^{\sqrt{\pi/2}} dx + \int_0^{\sqrt{\pi/2}} (-1) dx$$

$$= \sqrt{\frac{\pi}{2}} - 1 - \sqrt{\frac{\pi}{2}} = -1$$

$$\Rightarrow |I| = 1$$

Q5 (2)

$$f(x) = e^{-x} \sin x$$

$$\text{Now, } F(x) = \int_0^x f(t) dt \Rightarrow F'(x) = f(x)$$

$$I = \int_0^1 (F'(x) + f(x)) e^x dx = \int_0^1 (f(x) + f(x)) \cdot e^x dx$$

$$= 2 \int_0^1 f(x) \cdot e^x dx = 2 \int_0^1 e^{-x} \sin x \cdot e^x dx$$

$$= 2 \int_0^1 \sin x dx$$

$$= 2(1 - \cos 1)$$

$$I = 2 \left\{ 1 - \left(1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{6} + \frac{1}{8} - \dots \right) \right\}$$

$$I = 1 - \frac{2}{4} + \frac{2}{16} - \frac{2}{9} + \dots$$

$$1 - \frac{2}{4} < I < 1 - \frac{2}{4} + \frac{2}{6}$$

$$\frac{11}{12} < I < \frac{331}{360}$$

$$\Rightarrow I \in \left[\frac{11}{12}, \frac{331}{360} \right]$$

$$\Rightarrow I \in \left[\frac{330}{360}, \frac{331}{360} \right]$$

Q6 (1)

$$\text{Let } I = \int_0^{10} \frac{[\sin 2\pi x]}{e^{x-[x]}} dx = \int_0^{10} \frac{[\sin 2\pi x]}{e^{(x)}} dx$$

$$\text{Function } f(x) = \frac{[\sin 2\pi x]}{e^{[x]}} \text{ is periodic with}$$

period '1' Therefore

Hints and Solutions

MathonGo

$$\begin{aligned}
 I &= 10 \int_0^1 \frac{[\sin 2\pi x]}{e^{|x|}} dx \\
 &= 10 \int_0^1 \frac{[\sin 2\pi x]}{e^x} dx \\
 &= 10 \left(\int_0^{1/2} \frac{[\sin 2\pi x]}{e^x} dx + \int_{1/2}^1 \frac{[\sin 2\pi x]}{e^x} dx \right) \\
 &= 10 \left(0 + \int_{1/2}^1 \frac{(-1)}{e^x} dx \right) \\
 &= -10 \int_{1/2}^1 e^{-x} dx \\
 &= 10 (e^{-1} - e^{-1/2})
 \end{aligned}$$

Now,

$$10 \cdot e^{-1} - 10 \cdot e^{-1/2} = \alpha e^{-1} + \beta e^{-1/2} + \gamma \text{ (given)}$$

$$\Rightarrow \alpha = 10, \beta = -10, \gamma = 0$$

$$\Rightarrow \alpha + \beta + \gamma = 0$$

Q7 (1)

Let $\vec{x} = \lambda \vec{a} + \mu \vec{b}$ (λ and μ are scalars)

$$\vec{x} = \hat{i}(2\lambda + \mu) + \hat{j}(2\mu - \lambda) + \hat{k}(\lambda - \mu)$$

$$\text{Since } \vec{x} \cdot (3\hat{i} + 2\hat{j} - \hat{k}) = 0$$

$$3\lambda + 8\mu = 0$$

Also Projection of $\vec{x}$ on $\vec{a}$ is $\frac{17\sqrt{6}}{2}$

$$\frac{\vec{x} \cdot \vec{a}}{|\vec{a}|} = \frac{17\sqrt{6}}{2}$$

$$6\lambda - \mu = 51$$

$$\lambda = 8, \mu = -3$$

$$\text{From (1) and (2)} \quad \vec{x} = 13\hat{i} - 14\hat{j} + 11\hat{k}$$

$$|\vec{x}|^2 = 486$$

Q8 (512)

$$\begin{aligned} I &= 2 \int_0^4 f(x^2) dx \{ \text{Even function} \} \\ &= 2 \int_0^4 (4x^3 - g(4-x)) dx \\ &= 2 \left(\frac{4x^4}{4} \Big|_0^4 - \int_0^4 g(4-x) dx \right) \\ &= 2(256 - 0) = 512 \end{aligned}$$

Q9 (3)

$$\begin{aligned} \frac{1}{3} &\leq f(t) \leq 1 \forall t \in [0, 1] \\ 0 &\leq f(t) \leq \frac{1}{2} \forall t \in (1, 3] \\ \text{Now, } g(3) &= \int_0^3 f(t) dt = \int_0^1 f(t) dt + \int_1^3 f(t) dt \\ \therefore \int_0^1 \frac{1}{3} dt &\leq \int_0^1 f(t) dt \leq \int_0^1 1 dt \quad \dots (1) \\ \text{and } \int_1^3 0 dt &\leq \int_1^3 f(t) dt \leq \int_1^3 \frac{1}{2} dt \quad \dots (2) \\ \text{Adding, we get} \end{aligned}$$

$$\begin{aligned} \frac{1}{3} + 0 &\leq g(3) \leq 1 + \frac{1}{2}(3-1) \\ \frac{1}{3} &\leq g(3) \leq 2 \end{aligned}$$